Worcester Polytechnic Institute

RBE 595 Special Topics: Advanced Robot Navigation
Robotics Engineering
Spring 2024

Instructor:
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Communication:
For effective communication within the course, I encourage all students to utilize the Canvas discussion boards as the primary platform for asking questions, seeking clarifications, and engaging in course-related discussions. The discussion boards foster a collaborative learning environment where questions and their corresponding answers are accessible to the entire class, benefiting all students. This approach ensures that insights and solutions are shared widely, promoting a collective understanding of the course material. Please refrain from sending course-related inquiries directly via email or private messages unless it pertains to a personal matter.

Course Description:
In recent years, robots have become part of our everyday lives. Leaving the research labs to be part of the common tools of a household, tools such as robotic vacuum cleaners (iRobot Roomba, Kalorik), pool cleaners (Polaris, Maytronics), Lawn mowers (Landroid, LawnBott) and more abound. For navigating safely, these robots need the ability to localize themselves autonomously using their onboard sensors.

Potential applications of such systems include the automatic 3D reconstruction, 3D reconstruction of buildings, inspection and simple maintenance tasks, metric exploitation, surveillance of public places as well as in search and rescue systems. In this course, we will dive deep into the current techniques for 3D localization, mapping and navigation that are suitable for robotic applications.

Prerequisite Courses:
RBE 500 Foundations of Robotics; RBE 501 Robot Dynamics; RBE 502 Robot Control

Learning Outcomes:
The overall learning objective of this course is to develop an understanding of the mathematical foundations of modern probabilistic navigation techniques and to implement these methods in a suitable modern programming language such as Python or C++. By the end of this course a student should be able to:

1. Demonstrate comprehension of the fundamental mathematical theory of probabilistic methods for robotic navigation by implementing navigation techniques that draw from Bayesian filtering, Monte Carlo methods, and Markov models in a modern programming language.

2. Write sensor models, propagation models, and data fusion methods to properly incorporate information from various sensor components into a dynamic model of the robot.
3. Analyze and evaluate the performance of various navigation algorithms and techniques by metrics such as accuracy, precision, certainty, and computational efficiency.
4. Apply probabilistic navigation techniques to a wide scope of situations ranging from indoor two-dimensional problems up to global three-dimensional strategic navigation.

Textbook (and/or other Required Materials):
All required course content will be included in lectures and their accompanying slides and delivered online via Canvas. Any required readings will be provided. However, supplemental readings will be assigned from the following textbooks:

- *Probabilistic Robotics* by Sebastian Thrun, Wolfram Burgard, and Dieter Fox
- *Springer Handbook of Robotics* by Bruno Siciliano and Oussama Khatib
- *Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems – 2nd Edition* by Paul D. Groves

While these references will not be required for completing the course, they are exceptionally good references written and edited by some of the leading contributors to robotics and autonomous navigation over the past twenty years and are not particularly expensive books. Even better, *Springer Handbook of Robotics* can be downloaded as a PDF for free with your WPI credentials.

The course material will largely pull from *Probabilistic Robotics;* however, I will try my best to provide relevant equivalent readings in *Handbook of Robotics*. The assigned readings from *Principles of GNSS* will be provided.

Course Approach:
This course will run on a Monday to Sunday schedule. WPI (and myself) are based in Eastern Time (UTC -5/4 depending on daylight savings). The course is structured into several modules that culminate in a final module with a comprehensive project. These projects will be the main deliverables for the course. There will be some supplemental portions of your final grade that come from quizzes, discussions, et cetera, but this project-based curriculum is more reflective of actual professional work in the field. Course content will be conveyed using recorded lectures with accompanying slides, guided demonstrations in code, suggested weekly readings, and weekly discussions. I will be available for synchronous discussions on an as needed basis.

Course Requirements:

1. **Grade Determination Breakdown**
   This course will be primarily centered around projects. Each module will culminate in a module project which will enable the student to demonstrate knowledge and understanding of the module by applying it. The final module will also act as a culminating “final” project that incorporates everything from the semester.

2. **Assignments**
   There are five module projects with their respective grading weight shown below.
   - Bayes Filter 15%
   - Kalman Filter 15%
   - Nonlinear Kalman Filter 20%
   - Particle Filter 20%
   - INS/GNSS Integration 25%
   - Quizzes and participation 5%
3. **Late Work Policy**
   The modules are meant to be completed in order. The due dates on the module project are suggested completion dates there to help keep students on schedule. However, all projects are due at the end of the semester (a final deadline will be provided to allow for sufficient time to submit final grades) and the submissions are not capped provided students do not abuse this policy. Quizzes and discussion participation however will be due on a weekly basis.
4. **Class Participation Expectations and Criteria**
   There will be a discussion forum for each module, a weekly discussion on a specific topic, and a discussion forum for each assignment. The module and assignment discussions are there to encourage students to ask questions and work through problems they may have. Participation in the weekly discussions is expected. Discussion in the module and project forums is encouraged, but not required.

Technical Requirements:
Programming will be a major component of this course. Each project will require you to submit your raw code as well as an accompanying write-up detailing your results and the approach you took with your code including code snippets and details in the write up where necessary.

Students are free to select the language that they are most comfortable working with and the method for preparing the accompanying document. Generally, the lecture material will present language-agnostic pseudocode. However, any demonstrations will be given in Python and as such it is *strongly recommended* that students work in Python. Further, it is suggested that students develop their assignments using Jupyter Notebooks. The Anaconda Python distribution is recommended.

Library Access:
As a student at WPI, you have access to a variety of resources through the library. Use the link here to access databases, e-journals, and/or e-books. You will be required to log in with your WPI username and password to access materials.

POLICIES

Academic Integrity:

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You are expected to be familiar with the *Student Guide to Academic Integrity at WPI* that is downloadable from here. Consequences for violating the Academic Honest Policy range from earning a zero on the assignment, failing the course, or being suspended or expelled from WPI. Common examples of violations include:

- Copying and pasting text directly from a source without providing appropriately cited credit
- Paraphrasing, summarizing, or rephrasing from a source without providing appropriate citations
- Collaborating on individual assignments
- Turning in work where a good portion of the work is someone else's, even if properly cited

Academic Accommodations:
Courses should be compliant with the **American Disability Act**. Provide the following statement or a similar variation. If you, as the instructor, have questions about making content accessible to all learners, reach out to the Office of Accessibility Services.
We at WPI strive to create an inclusive environment where all students are valued members of the class community.

If you need course adaptations or accommodations because of a disability, or if you have medical information to share with us that may impact your performance or participation in this course, please make an appointment with us as soon as possible.
Students with approved academic accommodations should plan to submit their accommodation letters through the Office of Accessibility Services Student Portal. Should you have any questions about how accommodations can be implemented in this particular course, please contact me as soon as possible.

Students who are not currently registered with the Office of Accessibility Services (OAS) but who would like to find out more information regarding requesting accommodations and what that entails should plan to contact them via:
Email: AccessibilityServices@wpi.edu and/or
Phone: (508) 831-4908.
On Campus – Unity Hall

Mental Health
WPI's Student Development & Counseling Center (SDCC) staff are available for students in need of support: contact them directly and confidentially by phone at 508-831-5540 or email at SDCC@wpi.edu; or stop by their office at 16 Einhorn Road.

The I'm Concerned About a Student form provides a venue to connect with WPI services if you know of a student who is struggling and needs outreach.

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The **National Suicide Prevention Lifeline is a free, 24/7** service that can provide individuals with support, information, and local resources regarding suicide. **Call them at 1-800-273-TALK (8255) or visit www.suicidepreventionlifeline.org.** There is also Suicide Awareness & Prevention information on the WPI website.

If there is a chance someone is in imminent danger of suicidal action, WPI Campus Police can be reached at 508-831-5555 (or call 911 if off campus).

Grading Policy:
Final course grades are based on a student's performance as follows:

Letter Grade	Percentage
A	90 - 100
B	80 - 89
C	70 - 79
D	60 - 69
F	< 60

Course incompletes may be granted if the major part of the course is completed; however, no additional credit can be given for missed class discussions or teamwork beyond the end of the course. In addition, in the case of an incomplete, the student is responsible for handing in the final work within the WPI required timeframe of one (1) year. After this time, an incomplete grade changes to a failing (F) grade.

My philosophy on grading is centered around the Robotics Engineering PhD Technical qualifier where an average grade of roughly a B+ is used to demonstrate expertise and pass the qualifier. Similarly, I 'grade' assignments based on the following ratings: Excellent, Acceptable, Marginal, and Unacceptable, with details of what qualifications each level requires. I generally try to structure assignments such that an Acceptable rating corresponds to a B.

Course Schedule

Week #	**Topics**	**Readings**	**Assignments**
Week 1	Class introductions	PR 1.1-1.3	
Week 2	Bayes Filter	PR 2.1-2.5 SHR 4.3-4.4, 25.1.1-25.1.2	
Week 3	Kalman Filter	PR 3.1-3.2, 7.1-7.3 SHR 25.1.3	Bayes Filter
Week 4	Extended Kalman Filter	PR 3.3, 7.4-7.5 SHR 25.1.3	
Week 5	Unscented Kalman Filter	PR 3.4, 7.7-7.8	Kalman Filter
Week 6	Nonparametric Filters	PR 4.1-4.3	
Week 7	Monte Carlo Localization	PR 8.1-8.3 SHR 25.1.4	

Week 8	Coordinate Frames, Kinematics, and the Earth	Groves 2	Nonlinear KF
BREAK WEEK			
Week 9	Inertial Sensors	Groves 4 SHR 20	
Week 10	Inertial Navigation	Groves 5	Particle Filter
Week 11	Error State Kalman Filter		
Week 12	INS/GNSS Integration	Groves 14.1-14.3	
Week 13	INS/GNSS Integration	Groves 14.1-14.3	
Week 14	Bigger picture		
Week 15	Bigger picture		INS/GNSS